

TASK 4

Setup and Use a Firewall on Windows

Cybersecurity Internship Practical Report

Student	Cybersecurity Internship Trainee
Platform	Windows 10/11 (64-bit)
Security Tool	Windows Defender Firewall / netsh
Port Tested	TCP Port 23 (Telnet)
Practical Outcome	Rule created, verified, tested and removed
Report Date	4 September 2026

Purpose of this report

This report documents the practical work completed for Task 4. The exercise used Windows Firewall from an Administrator Command Prompt to inspect firewall profiles, create a temporary rule that blocks inbound TCP traffic on port 23, verify the rule, test connectivity, and finally remove the temporary rule.

Note: The connection test returned `TcpTestSucceeded : False`. This shows that a TCP connection to localhost port 23 was unsuccessful while the test was performed. The result is consistent with the port being blocked or otherwise not accepting connections; the test alone does not prove that the firewall was the only cause.

1. Objective

The objective was to understand basic host-based firewall management and network traffic filtering on Windows. A firewall rule was created to block inbound TCP traffic on port 23, the rule was inspected, connectivity was tested, and the temporary rule was removed after testing.

2. Technical Terms and Definitions

Term	Beginner-friendly definition
Firewall	A security control that monitors and filters network traffic according to rules. It can allow or block traffic.
Windows Firewall	The built-in Windows firewall used to control network traffic entering or leaving the computer.
Inbound traffic	Network traffic coming into the computer from another system or source.
Outbound traffic	Network traffic leaving the computer toward another system or service.
Port	A logical communication endpoint used by network applications. Different services commonly use different ports.
TCP	Transmission Control Protocol. A connection-oriented protocol that provides reliable delivery of data.
Port 23	The traditional port associated with Telnet.
Telnet	An older remote-login protocol. Traditional Telnet sends information without the encryption expected of modern protocols.
Firewall rule	A set of conditions that tells the firewall what traffic to allow or block.
Network profile	A Windows firewall context such as Domain, Private or Public, each with its own firewall configuration.
Loopback / 127.0.0.1	An address that refers back to the same computer. It is commonly used for local network testing.
netsh	Windows Network Shell, a command-line utility for viewing and configuring network-related settings.
PowerShell	A Windows command-line and scripting environment. In this task it was used to run Test-NetConnection.

3. Step-by-Step Procedure

Step 1 — Open Administrator Command Prompt

The commands were run from an elevated Command Prompt because changing firewall configuration requires administrative privileges.

Step 2 — Check the current firewall status

Before changing anything, the firewall status for all Windows network profiles was checked.

Step 3 — Create a temporary blocking rule

A rule named Block Telnet Port 23 was created for inbound TCP traffic on local port 23.

Step 4 — Verify the rule

The exact rule was displayed to confirm that it was enabled and configured to block TCP port 23.

Step 5 — Test the connection

A PowerShell Test-NetConnection command was used against 127.0.0.1 on port 23.

Step 6 — Remove the temporary rule

After testing, the rule was deleted so the system was returned to its previous firewall-rule state.

4. Commands Used and Explanation

A. Check all firewall profiles

```
netsh advfirewall show allprofiles
```

- netsh — Windows Network Shell.
- advfirewall — accesses the Advanced Firewall command set.
- show — displays information.
- allprofiles — displays settings for Domain, Private and Public profiles.
- Purpose: establish the starting firewall status before making a change.

B. Add the blocking rule

```
netsh advfirewall firewall add rule name="Block Telnet Port 23" dir=in action=block protocol=TCP localport=23
```

- add rule — creates a new firewall rule.
- name=... — gives the rule a readable name.
- dir=in — applies to inbound traffic.
- action=block — tells the firewall to block matching traffic.
- protocol=TCP — matches TCP traffic.
- localport=23 — matches local TCP port 23.
- Purpose: create the temporary security control required by the practical.

C. Show the created rule

```
netsh advfirewall firewall show rule name="Block Telnet Port 23"
```

- show rule — displays the selected rule's settings.
- Purpose: verify that the rule exists and that the direction, protocol, port and action are correct.

D. Test TCP port 23

```
powershell -Command "Test-NetConnection 127.0.0.1 -Port 23"
```

- powershell — launches PowerShell from Command Prompt.
- Test-NetConnection — tests network connectivity.
- 127.0.0.1 — loopback address for the same computer.
- -Port 23 — tests TCP port 23.
- Purpose: check whether a TCP connection to the local port can be established.

E. Delete the temporary rule

```
netsh advfirewall firewall delete rule name="Block Telnet Port 23"
```

- delete rule — removes the named firewall rule.
- Purpose: clean up the temporary lab configuration after testing.

5. Execution Evidence and Results

The following screenshots were captured during the practical. They are arranged in the same order as the procedure so an evaluator can follow the work from initial configuration through cleanup.

Evidence 1

```
C:\Windows\System32>netsh advfirewall show allprofiles

Domain Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                   N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096

Private Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                   N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096

Public Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                   N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
```

Figure 1 — Windows Firewall status for Domain, Private and Public profiles.

Evidence 2

```
C:\Windows\System32>netsh advfirewall firewall add rule name="Block Telnet Port 23" dir=in action=block protocol=TCP localport=23  
Ok.
```

Figure 2 — Firewall rule created successfully.

Evidence 3

```
C:\Windows\System32>netsh advfirewall firewall show rule name="Block Telnet Port 23"

Rule Name:                                Block Telnet Port 23
-----
Enabled:                                  Yes
Direction:                               In
Profiles:                                Domain,Private,Public
Grouping:
LocalIP:                                  Any
RemoteIP:                                 Any
Protocol:                                 TCP
LocalPort:                                23
RemotePort:                               Any
Edge traversal:                            No
Action:                                    Block
Ok.
```

Figure 3 — Verification of the Block Telnet Port 23 rule.

Evidence 4

RemoteManagement Disable

Test-NetConnection :: 127.0.0.1

Ping/ICMP Test

Waiting for echo reply

Figure 4 — TCP connection test to local port 23 failed.

Evidence 5

```
C:\Windows\System32>netsh advfirewall firewall delete rule name="Block Telnet Port 23"  
Deleted 1 rule(s).  
Ok.
```

Figure 5 — Temporary firewall rule deleted successfully.

6. Results and Interpretation

Activity	Observed result	Meaning
Firewall status	Domain, Private and Public profiles were ON	Windows Firewall was active across the displayed profiles.
Rule creation	Command returned "Ok."	The temporary firewall rule was created successfully.
Rule verification	Enabled: Yes; Direction: In; Protocol: TCP; Local Port: 23; Action: Block	The rule was added to the blocked configuration.
Connection test	TcpTestSucceeded: False.	The TCP connection to localhost port 23 was unsuccessful.
Cleanup	Deleted 1 rule(s). Ok.	The temporary rule was successfully removed.

7. What I Learned

- How to inspect Windows Firewall profiles from the command line.
- How a firewall rule uses direction, protocol, port and action to control traffic.
- How TCP ports relate to network services and connectivity testing.
- How to verify a firewall rule rather than assuming it was created correctly.
- How to use PowerShell's Test-NetConnection for a basic connectivity check.
- Why temporary lab rules should be removed after testing.

8. Security Significance

Firewalls are an important layer of endpoint security because they can reduce unwanted network access by applying rules to traffic. In this practical, port 23 was selected because it is associated with Telnet, an older remote-access protocol. Blocking unnecessary or insecure services is a common hardening principle, while modern secure administration should use appropriately protected protocols.

9. Conclusion

Task 4 was completed by checking the Windows Firewall configuration, creating and verifying a temporary inbound TCP port 23 blocking rule, testing the connection, and deleting the rule after the exercise. The practical provided hands-on experience with Windows Firewall, netsh, firewall rules, TCP ports and basic network connectivity testing.

Final status: TASK 4 PRACTICAL COMPLETED ✓